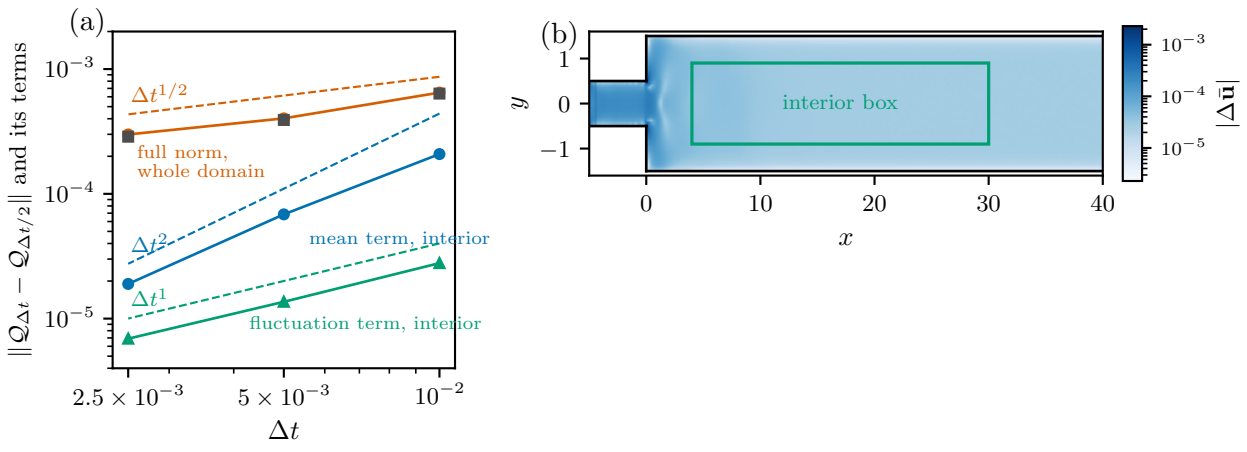

(a)
$\|\mathcal{Q}_{\Delta t} - \mathcal{Q}_{\Delta t/2}\|$ and its terms
10^{-3}
10^{-4}
10^{-5}
$\Delta t^{1/2}$
full norm,
whole domain
Δt^2
mean term, interior
Δt^1
fluctuation term, interior
2.5×10^{-3}
5×10^{-3}
10^{-2}
Δt
(b)
y
1
0
−1
interior box
0
10
20
30
40
x
10^{-3}
10^{-4}
10^{-5}
$|\Delta \bar{\mathbf{u}}|$